INTERNET OF THINGS · CHART 1

NodeMCU ESP8266

Fish Quality Monitor

Keeping the catch fresh on a fishing boat - watched from the owner's phone

Ice · Meltwater · Spoilage gas → box display + phone dashboard + voyage log

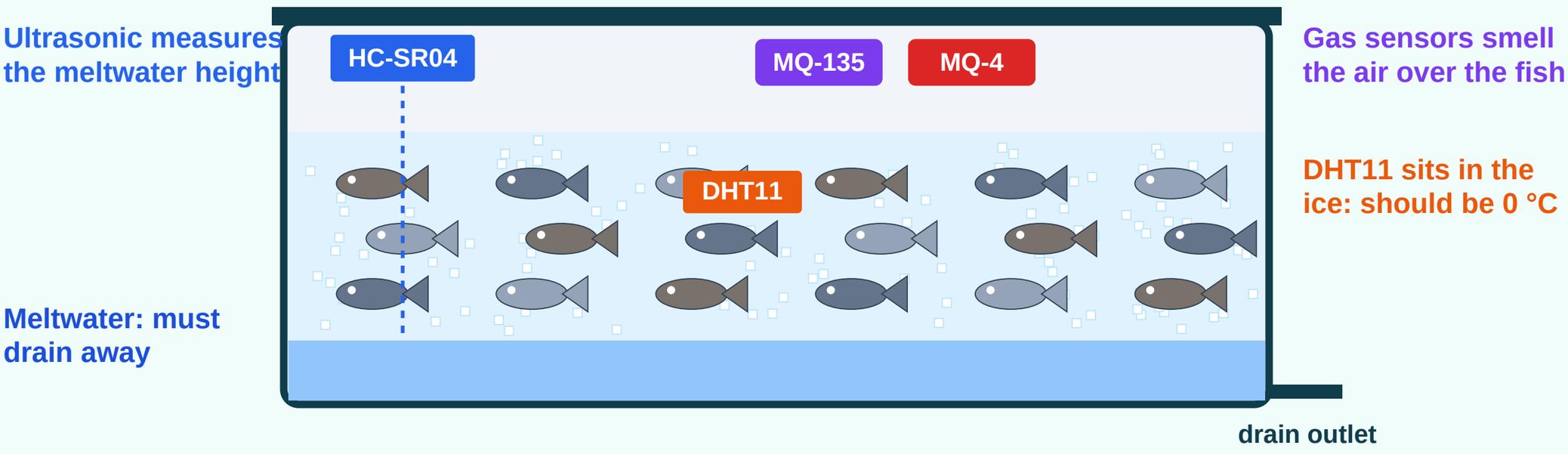
1 The problem

- The catch sits in a closed hold full of crushed ice.
- Ice melts, drains block, fish rot - unseen.
- The crew finds out only at the harbour.
- By then the catch has lost its price.

2 Our idea

- Sensors watch the hold **all the time** and log it.
- A box display shows a summary, **flashes** on trouble.
- The owner sees it all on a **phone or laptop**.
- No internet: the board makes its own WiFi.

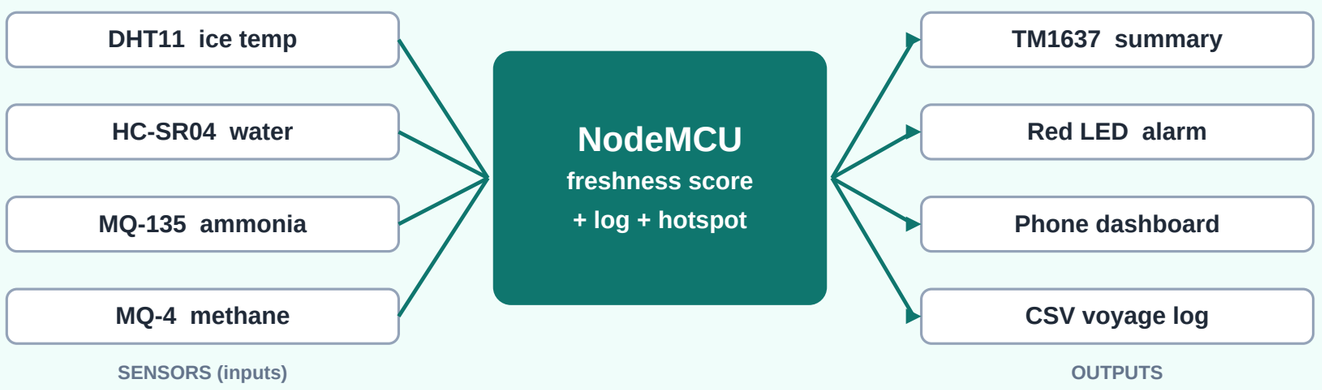
3 Inside the fish hold



4 Three dangers, four sensors

What goes wrong	Why it matters	Sensor
Ice runs out	above 0 °C bacteria multiply fast	DHT11
Drain outlet blocks	fish soak in dirty meltwater	HC-SR04
Fish start to rot	ammonia, amines ('fishy' smell)	MQ-135
Deep decay	methane gas forms	MQ-4

5 Parts we used

Part	Job	Rs.
NodeMCU ESP8266	brain, log, WiFi	280
TM1637 display	summary, fault codes	90
DHT11	ice temperature	110
HC-SR04	meltwater height	90
MQ-135	ammonia / amines	160
MQ-4	methane	160
Red LED, resistors	alarm light	10
5 V supply	power	150
Total		1,050

6 How the parts connect